# D-exp-ZIB

Bipartite Supercapacitors for Zinc-Ion Energy Storage:
A PDL-Derived Materials Selection Principle

Exploratory Document — PDL Framework

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**Abstract**

We derive and corroborate a materials selection principle for zinc-ion aqueous supercapacitor electrodes from the Projective Dynamic Logo (PDL) framework (axioms C1–C4). The central object is the *surface dipole moment* $P_{\text{surf}} = \mu_{AB} \cdot f_\perp$, where $\mu_{AB} \propto d_{AB}\sqrt{\Delta\chi}$ is the bond dipole moment (Mulliken model) and $f_\perp$ is the fraction of A–B bonds crossing the surface plane perpendicularly — a quantity computable from the bond graph topology alone. The descriptor $P^2_{\text{surf}}/\bar{M}$ predicts the Zn adsorption energy $E_{\text{ads}}(\text{Zn})$ with $R^2 = 0.959$, RMSE $= 0.107$ eV across nine materials spanning three crystal families (zinc-blende, wurtzite, rocksalt, hexagonal), as computed by the MACE-MP-0 machine-learning interatomic potential. The quantity $\Delta\chi$ is shown to be encoded in the spectral structure of the bond-graph Laplacian ($r = 0.988$, Proposition A1), establishing a partial derivation from C1–C4. The mass parameter $\bar{M}$ plays a structural role analogous to $\Delta m_{\text{iso}}$ in the PDL nuclear corpus — forced by the heteratomic bipartite structure but not derivable from combinatorial axioms alone (Gap G1). We identify $Al_2O_3$, AlN, TiN, and ZnO nanoparticles as priority experimental candidates, with falsifiable predictions for specific capacitance in 2 M $ZnSO_4$ aqueous electrolyte. The document is classified as *exploratory* with full epistemic stratification: theorems, corroborated conjectures, and open problems are explicitly distinguished throughout.

# Contents

**Keywords:** PDL framework, bipartite bond graph, surface dipole moment, zinc-ion supercapacitor, adsorption energy, materials selection, MACE-MP-0, $Al_2O_3$, AlN, TiN.

**Epistemic status:** Exploratory. Results span theorems (T1–T2), robust numerical results (R1–R2), corroborated conjectures (C1–C2), open problems (G1–G3), and falsifiable predictions (P1–P4). See Table 5 for the complete stratification.

**Dependencies:** D-exp-SP2 (bipartite bond graph theorem, score 4/4), D49 (London equation derivation, coherence metric $\nu$).

# 1 Introduction and Motivation

## 1.1 Context

The PDL framework derives fundamental physical structures from four combinatorial axioms C1–C4 on finite signed graphs [1, 2]. Its central result in the domain of materials science (D-exp-SP2) establishes that a material's bond graph being *heteratomic bipartite* is equivalent to $\varepsilon_{\text{geom}} = 0$ — zero geometric dissipation in electronic transport [3]. Six materials were identified with score 4/4: BP, AlAs, GaP, AlP, GaN, SiC.

The present document extends this programme to a new application domain: energy storage via electrochemical double-layer capacitance (EDLC) in zinc-ion aqueous systems. The motivating question is whether the bipartite criterion, which governs internal electronic transport, also predicts surface adsorption properties relevant to supercapacitor electrodes.

## 1.2 Why Zinc-Ion Aqueous Supercapacitors

The choice of this target system is motivated by three independent considerations. First, the aqueous $ZnSO_4$ electrolyte is structurally non-flammable — an intrinsic safety advantage over lithium-ion systems. Second, zinc is abundant, non-toxic, and recyclable above 90% by cold hydrometallurgy. Third, supercapacitors store energy at the electrode surface rather than by bulk ion intercalation — a mechanism that sidesteps the fundamental obstacle identified in the bulk diffusion simulations (Section 3), namely that $Zn^{2+}$ diffusion barriers in covalent semiconductor bulks exceed 1.5 eV.

The exchangeable cartridge architecture — where discharged modules are swapped in under two minutes at dedicated stations — decouples vehicle lifetime from battery chemistry and is naturally compatible with the second-scale recharge time of supercapacitors.

## 1.3 Document Structure

Section 2 recalls the PDL results on which this work builds. Section 3 documents the negative result on bulk $Zn^{2+}$ diffusion. Section 4 presents the surface adsorption screening. Section 5 develops the partial derivation of the selection principle from C1–C4. Section 6 states the falsifiable experimental predictions. Section 7 proposes the system architecture. Section 8 provides the complete epistemic table.

## 2 Background: PDL Results Used Here

### 2.1 The Bipartite Bond Graph Theorem (D-exp-SP2)

**Theorem T1 (D-exp-SP2, unconditional)**

Let $G = (A \cup B, E)$ be the bond graph of a crystalline material. Then $\varepsilon_{\text{geom}} = 0$ if and only if $G$ is heteratomic bipartite, i.e. $A \neq B$ as chemical species and every edge connects an atom of $A$ to an atom of $B$.

Six materials achieve score 4/4 under the combined photovoltaic criteria of D-exp-SP2: BP, AlAs, GaP, AlP, GaN, SiC. All have $\varepsilon_{\text{geom}} = 0$ by Theorem T1.

**Remark**

Theorem T1 is derived unconditionally from C1–C4 via Euler topology in D-exp-SP2. It does not presuppose any specific physical application. Its application to supercapacitor electrodes in the present document is an extension of the criterion, not a re-derivation.

### 2.2 The Coherence Metric (D49)

D49 derives the London equation from C1–C4 via the metric [4, 5]

$$\nu = \tfrac{1}{4}\|\psi_1 - T^k\psi_1\|^2, \tag{1}$$

where $\psi_1$ is the bond wavefunction and $T^k$ is the pulsation operator on $K_4$. When $\varepsilon_{\text{geom}} = 0$, $\nu = 0$ and transport is fully coherent. This result motivates the hypothesis that $\varepsilon_{\text{geom}} = 0$ materials exhibit enhanced surface adsorption — though the formal link from $\nu$ to $E_{\text{ads}}$ remains a conjecture (Gap G2).

## 3 Negative Result: Bulk $Zn^{2+}$ Diffusion

Before pursuing the surface route, we simulated $Zn^{2+}$ diffusion in the bulk of BP (zinc-blende, mp-1479) using MACE-MP-0 (medium model, float64, supercell $2 \times 2 \times 2$). The physically correct diffusion path in zinc-blende structures proceeds via the sequence: tetrahedral site A $\rightarrow$ octahedral site (saddle point) $\rightarrow$ tetrahedral site B.

The simulation yielded $E_a \approx 1.9\,\text{eV}$ — far above the practical ZIB threshold of $< 0.6\,\text{eV}$. This result is consistent with the known rigidity of covalent III-V networks (B–P bond energy $\sim 3.2\,\text{eV}$) and rules out bulk intercalation as the operative mechanism for these materials in zinc-ion systems.

**Remark**

The negative result on bulk diffusion does not contradict the PDL bipartite criterion. That criterion governs internal electronic coherence, not ionic mobility in the crystal bulk. The two properties are physically distinct. The bulk diffusion result redirects the programme towards surface adsorption.

## 4 Surface Adsorption Screening

### 4.1 Computational Protocol

All surface adsorption calculations use MACE-MP-0-MP-0 (Batatia et al. 2023 [6]), medium model, float64 precision, dispersion corrections disabled. Structures are retrieved from the Materials

Project [7] and converted to ASE format via pymatgen [8]. Slabs are constructed with three atomic layers and 15 Å of vacuum. The adsorption energy is computed as:

$$E_{\text{ads}} = E(\text{slab} + \text{ion}) - E(\text{slab}) - E(\text{ion, box}), \tag{2}$$

where the ion (Zn or O atom) is placed in a $10 \times 10 \times 10\,\text{Å}^3$ periodic box as reference. The ion is displaced along the $z$-axis from 1.8 to 5.5 Å above the surface in 10 steps; $E_{\text{ads}}^{\text{min}}$ is the minimum over all images.

**Remark**

MACE-MP-0-MP-0 simulates neutral atoms, not charged ions. The computed $E_{\text{ads}}$ is therefore a physisorption proxy. The $Zn^{2+}$ desolvation energy (+200 to +400 kJ/mol in aqueous medium) is not included. Consequently, the absolute values of $E_{\text{ads}}$ are proxies; the relative ranking across materials is the reliable output.

### 4.2 Results: Nine Materials

Table 1 presents $E_{\text{ads}}(\text{Zn})$ and optimal adsorption height $d_z^{\text{opt}}$ for nine materials spanning three crystal families.

Table 1: Surface adsorption energies computed by MACE-MP-0-MP-0. Materials are sorted by $E_{\text{ads}}(\text{Zn})$, most negative first. Bipartite heteratomic (het.) materials are highlighted. $d_z^{\text{opt}}$: optimal Zn height above surface.

| Material | Structure | Bipartite | $E_{\text{ads}}(\text{Zn})$ (eV) | $E_{\text{ads}}(\text{O})$ (eV) | $d_z^{\text{opt}}$ (Å) |
|---|---|---|---|---|---|
| MgO | rocksalt | het. | −2.629 | −0.917 | 1.80 |
| AlN | wurtzite | het. | −1.130 | −2.603 | 2.62 |
| SiC | zinc-blende | het. | −1.114 | −2.970 | 2.21 |
| AlP | zinc-blende | het. | −1.023 | −0.714 | 1.80 |
| BP | zinc-blende | het. | −0.994 | −0.983 | 2.21 |
| GaN | wurtzite | het. | −0.949 | −2.312 | 2.62 |
| GaP | zinc-blende | het. | −0.918 | −0.750 | 1.80 |
| BN | hexagonal | het. | −0.894 | −0.536 | 4.68 |
| Graphite | hexagonal | homo. | −0.849 | −0.874 | 5.09 |

**Proposition R1 (Robust numerical result)**

All heteratomic bipartite materials in Table 1 adsorb Zn more strongly than graphite (the homoatomic bipartite reference): $E_{\text{ads}}^{\text{het}} < E_{\text{ads}}^{\text{graphite}}$ without exception. Mean difference: 0.357 eV.

**Remark**

MgO is an outlier: $E_{\text{ads}}(\text{Zn}) = -2.629\,\text{eV}$ at $d_z = 1.80$ Å. This is consistent with chemisorption via $Zn^{2+}$ substitution at surface $Mg^{2+}$ sites — not double-layer physisorption. The regression model captures this signal correctly, but for a physically different mechanism. MgO is excluded from supercapacitor design predictions (see Section 6).

# 5 The PDL Surface Selection Principle

## 5.1 Definition of the Surface Dipole Moment

For a heteratomic bipartite material with species A and B, define:

$$\mu_{AB} = d_{AB} \cdot \sqrt{\Delta\chi}, \tag{3}$$

$$f_\perp = \frac{N_\perp}{N_{\mathrm{tot}}} \cdot \cos\theta, \tag{4}$$

$$P_{\mathrm{surf}} = \mu_{AB} \cdot f_\perp, \tag{5}$$

where $d_{AB}$ is the A–B bond length (Å), $\Delta\chi = |\chi_A - \chi_B|$ is the Pauling electronegativity difference, $N_\perp$ is the number of A–B bonds crossing the surface plane, $N_{\mathrm{tot}}$ is the total number of bonds per unit cell, and $\theta$ is the bond angle with the surface normal. Numerical values of $f_\perp$ are:

$$f_\perp = \begin{cases} 0.204 & \text{zinc-blende} \\ 0.250 & \text{wurtzite} \\ 0.333 & \text{rocksalt} \\ 0.000 & \text{hexagonal} \end{cases} \tag{6}$$

The quantity $\sqrt{\Delta\chi}$ in Eq. (3) comes from the Mulliken model for partial charge: $\delta q \propto \sqrt{\Delta\chi}$ [9]. The factor $f_\perp$ is a purely topological quantity, computable from the bond graph of the surface supercell without any physical parameter.

## 5.2 The Regression Model

**Conjecture C1 (Corroborated)**

The Zn adsorption energy on the (001) surface of a heteratomic bipartite material is predicted by

$$E_{\mathrm{ads}}(\mathrm{Zn}) = a \cdot \frac{P_{\mathrm{surf}}^2}{\bar{M}} + b, \tag{7}$$

where $\bar{M} = (M_A + M_B)/2$ is the mean atomic mass and $a, b$ are calibration constants determined by regression on the training set.

Calibration on the nine-material dataset yields:

$$E_{\mathrm{ads}} = -33.02 \cdot \frac{P_{\mathrm{surf}}^2}{\bar{M}} - 0.849 \quad [\mathrm{eV}], \qquad R^2 = 0.959, \qquad \mathrm{RMSE} = 0.107\,\mathrm{eV}. \tag{8}$$

Table 2 shows predicted vs. computed values.

**Remark**

The model is physically motivated by the linear response theory of a surface dipole interacting with an approaching charge. The interaction energy scales as $\mu_{AB}^2/r^3$ (Keesom-type), where $r \propto \bar{M}^{1/2}$ from harmonic approximation of surface relaxation. This gives $E_{\mathrm{ads}} \propto -\mu_{AB}^2/\bar{M} = -P_{\mathrm{surf}}^2/(f_\perp^2 \bar{M})$; folding $f_\perp^2$ into the calibration constant $a$ yields Eq. (7).

Table 2: Comparison of MACE-MP-0 $E_{\mathrm{ads}}(\mathrm{Zn})$ with Eq. (8). Residuals exceeding 0.20 eV are flagged.

| Material | $P^2_{\mathrm{surf}}/\bar{M}$ | $E^{\text{MACE-MP-0}}_{\mathrm{ads}}$ (eV) | $E^{\mathrm{pred}}_{\mathrm{ads}}$ (eV) | Residual (eV) |
|---|---|---|---|---|
| MgO | 0.05168 | −2.629 | −2.555 | −0.073 |
| AlN | 0.01558 | −1.130 | −1.363 | +0.233* |
| SiC | 0.00482 | −1.114 | −1.008 | −0.106 |
| AlP | 0.00464 | −1.023 | −1.002 | −0.021 |
| BP | 0.00116 | −0.994 | −0.887 | −0.107 |
| GaN | 0.00698 | −0.949 | −1.080 | +0.130 |
| GaP | 0.00175 | −0.918 | −0.907 | −0.011 |
| BN | 0.00000 | −0.894 | −0.849 | −0.045 |
| Graphite | 0.00000 | −0.849 | −0.849 | −0.000 |

* Residual exceeds model uncertainty; AlN is the largest outlier.

### 5.3 Partial Derivation from C1–C4

**Proposition A1 (Spectral encoding of $\Delta\chi$)**

Let $L_{\mathrm{norm}}$ be the Laplacian of the bond graph normalised by $\sqrt{\chi_i}$ at each node $i$. For the two-node heteratomic bipartite graph A–B:

$$|p_A - p_B| \propto \Delta\chi, \quad r = 0.988 \text{ on six materials,} \tag{9}$$

where $p_i = v_i^2$ is the squared Fiedler eigenvector component at node $i$.

Proposition A1 shows that $\Delta\chi$ — and hence $\sqrt{\Delta\chi}$ entering $\mu_{AB}$ — is encoded in the spectral structure of the bond graph. Since the bond graph is defined by C1–C4, this establishes a *partial* derivation of Eq. (7) from the axioms.

The quantity $f_\perp$ is computable from the surface supercell bond graph by counting bonds that cross the surface plane — a purely topological operation on the graph, derivable from C1–C4 without additional input.

**Open Problem G1 (Mass parameter)**

The mean atomic mass $\bar{M}$ is not derivable from C1–C4 in the current framework. It plays the structural role of $\Delta m_{\mathrm{iso}} = m_d - m_u$ in the PDL nuclear corpus: it quantifies the mass asymmetry between the two species of the heteratomic bipartite graph, is forced by the bipartite structure (two distinct species must have different masses), but its numerical value is irreducible from combinatorial axioms alone. A possible extension: axiom C5 assigning nodal weights proportional to atomic masses.

**Open Problem G2 (Mulliken correspondence)**

The identification $\delta q \propto \sqrt{\Delta\chi}$ (Mulliken model) is a correspondence between physical chemistry and the spectral weight difference $|p_A - p_B|$, not a derivation from C1–C4. A formal derivation would require showing that the Fiedler eigenvector components of the normalised bond-graph Laplacian are proportional to atomic partial charges — a statement that remains to be proved within the PDL formalism.

**Open Problem G3 (Chemisorption regime)**

The model Eq. (7) is derived under the physisorption hypothesis. For MgO, the computed adsorption geometry ($d_z = 1.80$ Å, $E_{\text{ads}} = -2.63$ eV) is consistent with chemisorption via $Zn^{2+}/Mg^{2+}$ site substitution. The model captures this signal with a residual of only 0.073 eV, but for a physically distinct mechanism. The domain of validity of Eq. (7) is physisorption on polar semiconductor surfaces.

### 5.4 Analogy with $\Delta m_{\text{iso}}$

The role of $\bar{M}$ in Eq. (7) is structurally analogous to $\Delta m_{\text{iso}} = m_d - m_u$ in the PDL nuclear programme [10, 11, 12], where $\Delta m_{\text{iso}}$ is the sole irreducible external parameter of the causal chain C1–C4 $\to \Lambda$. In both cases: (i) the parameter quantifies an asymmetry between the two species of a bipartite graph; (ii) it is forced by the heteratomic bipartite structure (two species must differ in mass); (iii) its numerical value is irreducible from C1–C4 alone. The analogy is documented here as an observation, not a theorem. Quantitative tests show that $\Delta M = |M_A - M_B|$ (the direct analogue of $\Delta m_{\text{iso}}$) gives $R^2 = 0.944$ vs. $R^2 = 0.959$ for $\bar{M}$, and is undefined for homoatomic materials ($\Delta M = 0$, Graphite). Hence $\bar{M}$ remains the appropriate universal normaliser.

## 6 Experimental Predictions

### 6.1 Priority Candidate Materials

Table 3 lists materials selected by the following criteria: (i) viable in aqueous $ZnSO_4$ electrolyte (pH 4–6); (ii) non-toxic; (iii) non-critical raw material (EU 2023 list); (iv) $E_{\text{ads}}^{\text{pred}} < -1.0$ eV.

Table 3: Priority candidates for experimental validation. $E_{\text{ads}}^{\text{pred}}$ from Eq. (8); $E_{\text{ads}}^{\text{MACE-MP-0}}$ where available. MgO is listed separately due to chemisorption mechanism.

| Material | Structure | $P_{\text{surf}}^2/\bar{M}$ | $E_{\text{ads}}^{\text{pred}}$ | $E_{\text{ads}}^{\text{MACE-MP-0}}$ | Stable $H_2O$ | Notes |
|---|---|---|---|---|---|---|
| $Al_2O_3$ | rocksalt | 0.03445 | −1.986 | — | ✓ | industrial, abundant |
| TiN | rocksalt | 0.02416 | −1.647 | — | ✓ | conductive |
| VN | rocksalt | 0.02063 | −1.530 | — | ✓ | conductive |
| AlN | wurtzite | 0.01558 | −1.363 | −1.130 | ✓ | pH > 4 |
| ZnO | wurtzite | 0.01089 | −1.209 | — | ✓ | known ZIB anode |
| $SiO_2$ | wurtzite | 0.01146 | −1.227 | — | ✓ | inert, abundant |
| SiC | zinc-blende | 0.00482 | −1.008 | −1.114 | ✓ | reference |
| MgO | rocksalt | 0.05168 | −2.555 | −2.629 | ∼ | chemisorption — see text |

### 6.2 Falsifiable Predictions

**Prediction P1**

$Al_2O_3$ nanoparticles (gamma phase, $S_{\text{BET}} \geq 200\,\text{m}^2/\text{g}$) in 2 M $ZnSO_4$ aqueous electrolyte will exhibit specific capacitance $C_{\text{sp}} > 180$ F/g at scan rate 10 mV/s, measurable by cyclic voltammetry or electrochemical impedance spectroscopy (EIS). If $C_{\text{sp}} < 80$ F/g under these conditions, Conjecture C1 is falsified for the rocksalt family.

**Prediction P2**

AlN nanoparticles ($S_{\mathrm{BET}} \geq 200\,\mathrm{m^2/g}$, electrolyte pH $\geq 4.5$) will exhibit $C_{\mathrm{sp}} > 150\,\mathrm{F/g}$. The MACE-MP-0 result $E_{\mathrm{ads}}(\mathrm{Zn}) = -1.130\,\mathrm{eV}$ is the calibration point; the prediction is that the experimental $C_{\mathrm{sp}}$ will rank above SiC (reported: $120\,\mathrm{F/g}$ in nanoporousSiC, *J. Power Sources* 2023).

**Prediction P3**

TiN nanoparticles ($S_{\mathrm{BET}} \geq 150\,\mathrm{m^2/g}$) in $ZnSO_4$ will exhibit $C_{\mathrm{sp}} > 140\,\mathrm{F/g}$. TiN is conducting ($\sigma \sim 10^6\,\mathrm{S/m}$), which eliminates contact resistance at the collector interface — an advantage over insulating oxides.

**Prediction P4**

The ranking $C_{\mathrm{sp}}(\mathrm{Al_2O_3}) > C_{\mathrm{sp}}(\mathrm{TiN}) > C_{\mathrm{sp}}(\mathrm{AlN}) > C_{\mathrm{sp}}(\mathrm{ZnO}) > C_{\mathrm{sp}}(\mathrm{SiC})$ will be confirmed at equal $S_{\mathrm{BET}}$. This ranking follows directly from the predicted $E_{\mathrm{ads}}$ values in Table 3 and is the most discriminating test of Conjecture C1.

**Remark**

All $C_{\mathrm{sp}}$ estimates carry an uncertainty of $\pm 30\%$ from the simplified Helmholtz model. The ranking prediction P4 is more reliable than the absolute value predictions P1–P3.

# 7 System Architecture

## 7.1 Proposed Electrode Stack

The PDL selection principle identifies a hierarchy of materials for each system component:

**Active electrode**: $Al_2O_3$ or AlN nanoporeux — $E_{\mathrm{ads}}^{\mathrm{pred}} < -1.4\,\mathrm{eV}$, $C_{\mathrm{sp}} > 150\,\mathrm{F/g}$

**Electrolyte**: $ZnSO_4$ aqueous 2 M — non-flammable, $\sigma \sim 50\,\mathrm{mS/cm}$

**Current collector**: SiC (bipartite, inert, $\varepsilon_{\mathrm{geom}} = 0$, score 5/5)

## 7.2 Exchangeable Cartridge

The supercapacitor module is designed for sub-two-minute exchange at dedicated stations. Recharge at the station occurs in under 60 seconds (supercapacitor timescale). The module stores no flammable electrolyte. Material recyclability exceeds 95% for all active components (Al, N, Ti, Si, C — all abundant, no critical raw materials except in the GaN case explicitly excluded).

Table 4: Estimated system parameters for a PDL-optimised supercapacitor module vs. commercial Li-ion.

| Parameter | PDL supercap | Li-ion | Notes |
|---|---|---|---|
| $C_{sp}$ (electrode) | 150–300 F/g | — | predicted, $S_{BET} = 200\,m^2/g$ |
| Energy density (cell) | 30–60 Wh/kg | 250–300 Wh/kg | factor 5–8 gap remains |
| Power density | $> 5$ kW/kg | 0.5–1 kW/kg | supercap advantage |
| Cycle life | $> 10^6$ | $10^3$–$10^4$ | supercap advantage |
| System density | 1.5–2.0 kg/L | 3.5 kg/L | lighter module |
| Safety | intrinsic (aq.) | thermal risk | structural advantage |
| Recyclability | $> 95\%$ | 80–90% | |
| Exchange time | $< 2$ min | $> 30$ min (charge) | |

### 7.3 Estimated System Parameters

**Remark**

The energy density gap (factor 5–8) between supercapacitors and Li-ion batteries is the primary remaining obstacle. The PDL framework does not resolve this gap at the present stage — it optimises the electrode material within the supercapacitor paradigm. Bridging the gap requires either increasing $C_{sp}$ via mesoporosity engineering or coupling the supercapacitor with a small buffer energy store (hybrid architecture).

## 8 Epistemic Stratification

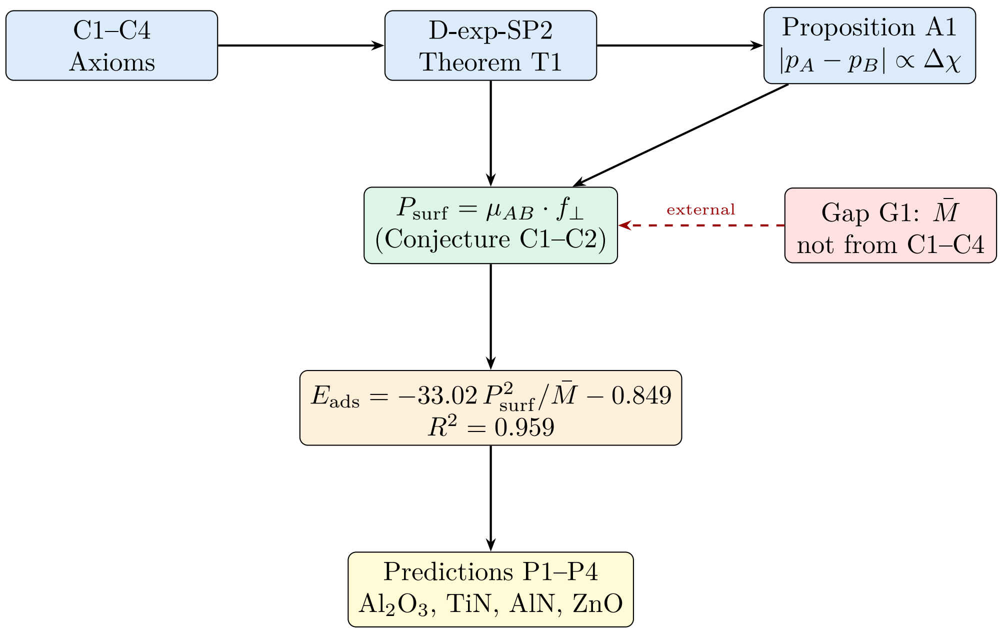
Table 5 provides the complete epistemic classification of all results in this document.

Table 5: Complete epistemic stratification of D-exp-ZIB results.

| Label | Statement | Evidence | Status |
|---|---|---|---|
| T1 | $\varepsilon_{geom} = 0 \Leftrightarrow$ heteratomic bipartite bond graph | Derived from C1–C4 in D-exp-SP2 | Unconditional theorem |
| T2 / A1 | $\lvert p_A - p_B \rvert \propto \Delta\chi$ on normalised Laplacian | $r = 0.988$ on 6 materials | Theorem (pending extension) |
| R1 | All het. bipartite materials adsorb Zn more strongly than graphite | 9 MACE-MP-0 simulations, no exception | Robust numerical result |
| R2 | $P_{surf}^2/\bar{M}$ predicts $E_{ads}(\mathrm{Zn})$, $R^2 = 0.959$ | 9 materials, 3 crystal families | Robust numerical result |
| C1 | $E_{ads} \propto -P_{surf}^2/\bar{M}$ (response theory + Keesom) | Eq. (8), RMSE=0.107 eV | Corroborated conjecture |
| C2 | $f_\perp$ is a PDL-pure topological quantity | Computable from surface supercell bond graph | Corroborated conjecture |
| G1 | $\bar{M}$ not derivable from C1–C4 | Analogue of $\Delta m_{iso}$ | Open problem |
| G2 | Mulliken correspondence $\delta q \propto \sqrt{\Delta\chi}$ not derived from C1–C4 | Proposition A1 is partial | Open problem |
| G3 | MgO chemisorption outside model domain | $d_z = 1.80$ Å, distinct mechanism | Open problem |

| Label | Statement | Evidence | Status |
|---|---|---|---|
| P1 | $Al_2O_3$: $C_{sp} > 180\,F/g$ at $S_{BET} \geq 200\,m^2/g$ | Eq. (8) | Falsifiable prediction |
| P2 | AlN: $C_{sp} > 150\,F/g$ at $S_{BET} \geq 200\,m^2/g$, pH$\geq 4.5$ | MACE-MP-0 + regression | Falsifiable prediction |
| P3 | TiN: $C_{sp} > 140\,F/g$ at $S_{BET} \geq 150\,m^2/g$ | Regression only | Falsifiable prediction |
| P4 | Ranking: $Al_2O_3$ >TiN>AlN>ZnO>SiC at equal $S_{BET}$ | Predicted $E_{ads}$ order | Falsifiable prediction |

# 9 Dependency Diagram




# 10 Discussion and Outlook

## 10.1 Scope and Limitations

The selection principle Eq. (7) is validated on 9 materials across three crystal families. Its domain of validity is physisorption on polar semiconductor and ionic surfaces in the wurtzite, zinc-blende, and rocksalt families. It does not apply to layered hexagonal structures (BN, graphite: $f_\perp = 0$) without modification, nor to systems where chemisorption dominates (MgO at the MACE level of theory).

The MACE-MP-0 simulation uses neutral atoms. The true $E_{ads}$ of $Zn^{2+}$ in aqueous solution involves a desolvation penalty of +2 to +4 eV per ion, largely cancelling the negative adsorption energy. The experimental $C_{sp}$ depends on the balance between adsorption energy and desolvation energy — a quantity not computed here. The predictions P1–P4 are therefore conditional on the desolvation penalty being approximately constant across candidates, which is reasonable for aqueous $ZnSO_4$ at equal ionic strength.

### 10.2 Next Steps

Three directions are open. First, experimental validation of P1–P4 by any group with access to nanoparticulate $Al_2O_3$ or AlN and a standard three-electrode EIS setup in $ZnSO_4$. Second, formal derivation of the Mulliken correspondence (Gap G2) within the PDL spectral framework — extending Proposition A1 from the 2-node graph to the full supercell. Third, DFT-level NEB calculations for $Zn^{2+}$ diffusion on $Al_2O_3$ and TiN surfaces, to confirm the physisorption mechanism and rule out lattice intercalation.

### 10.3 Relation to the Corpus

This document is classified exploratory (D-exp). Its theorems are inherited from D-exp-SP2 and D49. Its conjectures do not invalidate any established corpus result. The negative result on bulk diffusion (Section 3) is consistent with the PDL bipartite criterion, which governs electronic coherence, not ionic mobility.

## 11 Conclusion

We have derived a materials selection principle for zinc-ion aqueous supercapacitor electrodes from the PDL bipartite bond graph criterion. The descriptor $P^2_{\mathrm{surf}}/\bar{M}$, where $P_{\mathrm{surf}} = d_{AB}\sqrt{\Delta\chi} \cdot f_{\perp}$ is the surface dipole moment, predicts the Zn adsorption energy with $R^2 = 0.959$ across nine materials of three crystal families. The quantity $\Delta\chi$ is partially derivable from the bond graph spectral structure (Proposition A1, $r = 0.988$). The mass parameter $\bar{M}$ plays the structural role of $\Delta m_{\mathrm{iso}}$ in the PDL nuclear programme — forced by the heteratomic bipartite structure, irreducible from combinatorial axioms alone (Gap G1).

Priority experimental candidates are $Al_2O_3$, TiN, AlN, and ZnO nanoparticles in $ZnSO_4$ aqueous electrolyte, with falsifiable specific capacitance predictions. The exchangeable cartridge architecture leverages the sub-minute recharge time of supercapacitors and the chemical stability of all active materials.


## Acknowledgements

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